

CHEMICAL THERMODYNAMICS

1. $A + B \longrightarrow C + D$
 $\Delta H = -10,000 \text{ J mol}^{-1}$
 $\Delta S = -33.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 At what temperature the reaction will occur spontaneous from left to right?
 (1) = 300.3 K (2) > 300.3 K
 (3) < 300.3 K (4) None of these
2. $C_2H_4 + Cl_2 \longrightarrow C_2H_4Cl_2$
 $\Delta H = -270.6 \text{ KJ mol}^{-1}$
 $\Delta S = -139.0 \text{ J K}^{-1} \text{ mol}^{-1}$
 Find value of ΔG at $T = 300 \text{ K}$.
 (1) $-328.9 \text{ KJ mol}^{-1}$ (2) $-228.9 \text{ KJ mol}^{-1}$
 (3) $-129.8 \text{ KJ mol}^{-1}$ (4) $-528.9 \text{ KJ mol}^{-1}$
3. 0.16 g of methane was subjected to combustion at 27°C in a bomb calorimeter system. The temperature of the calorimeter system (including water) was found to rise by 0.5°C . Heat of combustion of methane at constant pressure is- (Heat capacity of the calorimeter system is 17.7 KJ K^{-1}).
 (1) -890 KJ (2) -885 KJ
 (3) $+890 \text{ KJ}$ (4) $+885 \text{ KJ}$
4. The standard heats of formation of $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ are -76.2 , -394.8 and $-241.6 \text{ KJ mol}^{-1}$ respectively. Heat evolved by burning 1m^3 of methane measured at NTP will be-
 (1) -801.8 KJ (2) -35794.6 KJ
 (3) -890 KJ (4) -3570 KJ
5. 10 moles of an ideal gas expand isothermally and reversibly from a pressure of 5 atm to 1 atm at 300K. what is work done-
 (1) $-40.15 \times 10^3 \text{ J}$ (2) $-20.30 \times 10^3 \text{ J}$
 (3) $-18 \times 10^3 \text{ J}$ (4) $-55 \times 10^3 \text{ J}$
6. Enthalpy of formation of H_2O is -68 Kcal/mole . Enthalpy of formation of OH^- ions is -
 $\left[\text{Given } \Delta H_f^\circ(H^+) = 0 \right]$
 (1) $-13.7 \text{ Kcal mol}^{-1}$ (2) $-57.3 \text{ Kcal mol}^{-1}$
 (3) $-54.3 \text{ Kcal mol}^{-1}$ (4) $-20.1 \text{ Kcal mol}^{-1}$
7. Heat released upon combustion of 2gm of C_{graphite} is X KJ. Heat of formation of CO_2 is-
 (1) $-6X \text{ KJ}$ (2) $-X \text{ KJ}$
 (3) $+6X \text{ KJ}$ (4) $+X \text{ KJ}$
8. A swimmer coming out from a pool is covered with a film of water weighing about 80g. How much heat must be supplied to evaporate this water ($\Delta_{\text{vap}} H^\circ = 40.79 \text{ KJ mol}^{-1}$) -
 (1) 181 KJ (2) 362 KJ
 (3) 90.5 KJ (4) 40.79 KJ
9. Calculate the entropy change in melting of one gm ice at 0°C if latent heat of ice is 80 cal/g
 (1) 80 Cal K^{-1} (2) 20 Cal K^{-1}
 (3) 4.4 Cal K^{-1} (4) 0.3 Cal K^{-1}
10. For the water gas reaction
 $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$
 The standard Gibbs's energy of reaction (at 1000 K) is -8.1 KJ mol^{-1} . Value of equilibrium constant is-
 (1) 2.6 (2) 6.2 (3) 8.2 (4) 10
11. Gibbs's energy change for the dissolution of potassium nitrate in water at 298 K is-
 $KNO_3(s) + \text{water} \longrightarrow K^+(aq) + NO_3^-(aq)$
 $\Delta H^\circ = 34 \text{ KJ mol}^{-1}$
 $\Delta S^\circ = 0.116 \text{ KJ K}^{-1} \text{ mol}^{-1}$
 (1) $+0.568 \text{ KJ mol}^{-1}$
 (2) $-0.568 \text{ KJ mol}^{-1}$
 (3) $-1.036 \text{ KJ mol}^{-1}$
 (4) $+1.036 \text{ KJ mol}^{-1}$
12. Determine the entropy change for the reaction given below:
 $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$ at 300 K.
 If standard entropies of $H_2(g)$, $O_2(g)$ and $H_2O(l)$ are 126.6, 201.20 and $68.0 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively.
 (1) $-218.4 \text{ J K}^{-1} \text{ mol}^{-1}$
 (2) $-318.4 \text{ J K}^{-1} \text{ mol}^{-1}$
 (3) $-520.2 \text{ J K}^{-1} \text{ mol}^{-1}$
 (4) $-128.6 \text{ J K}^{-1} \text{ mol}^{-1}$
13. Out of E, H, q, W and S which are state function?
 (1) E, H, W (2) E, S, H, W
 (3) E, H, S (4) E, H, q, W, S
14. Bond energy of C - H bond is-
 Given that the heat of formation of CH_4 , heat of sublimation of carbon and heat of dissociation of H_2 are -74.8 , $+719.6$ and $435.4 \text{ KJ mol}^{-1}$ respectively?
 (1) 1665.2 KJ (2) 416.3 KJ
 (3) 816.3 KJ (4) 614.2 KJ

- 15.** Energy required to raise the temperature of 10.0 g of iron from 25°C to 500°C, if specific heat capacity of iron is $0.45 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$ is—
 (1) 3020.2 J (2) 2137.5 J
 (3) 1537 J (4) 836 J
- 16.** Calculate the Gibb's energy change when 1 mole of NaCl is dissolved in water at 25°C.
 Lattice energy of NaCl = $777.8 \text{ KJ mol}^{-1}$;
 ΔS for dissolution = $0.043 \text{ KJ mol}^{-1}$ and
 Hydration energy of NaCl = $-774.1 \text{ KJ mol}^{-1}$
 (1) $-9.114 \text{ KJ mol}^{-1}$ (2) $-11.4 \text{ KJ mol}^{-1}$
 (3) -5.4 KJ mol^{-1} (4) -4.5 KJ mol^{-1}
- 17.** In which reactions ΔS is positive
 (1) $\text{H}_2\text{O}(\ell) \longrightarrow \text{H}_2\text{O}(\text{s})$
 (2) $3\text{O}_2(\text{g}) \longrightarrow 2\text{O}_3(\text{g})$
 (3) $\text{H}_2\text{O}(\ell) \longrightarrow \text{H}_2\text{O}(\text{g})$
 (4) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
- 18.** When enthalpy and entropy change for a chemical reaction are $-2.5 \times 10^3 \text{ cal.}$ and $7.4 \text{ cal. deg}^{-1}$ respectively predict the reaction at 298 K is—
 (1) Spontaneous (2) Non spontaneous
 (3) At equilibrium (4) None of these
- 19.** If a gas at constant temperature and pressure expands then its—
 (1) Internal energy increases
 (2) Internal energy remains the same
 (3) Internal energy decreases
 (4) Entropy increases and then decreases
- 20.** A 1.250 g sample of octane (C_8H_{18}) is burned in excess of oxygen in a bomb calorimeter. The temperature of the calorimeter rises from 294.05K to 300.78 K. If heat capacity of the calorimeter is 8.93 KJ/K , enthalpy of combustion of the sample of octane at constant volume is—
 (1) $-6285 \text{ KJ mol}^{-1}$ (2) $-5481.1 \text{ KJ mol}^{-1}$
 (3) $-2800 \text{ KJ mol}^{-1}$ (4) $-3680 \text{ KJ mol}^{-1}$
- 21.** Which has maximum internal energy?
 (1) O_3 gas (2) O_2 gas
 (3) He gas (4) Both (2) and (3)
- 22.** Which is not true about G?
 (1) If $\Delta G > 0$, reaction is spontaneous
 (2) G is a state function
 (3) $-\Delta G = W_{\text{useful}}$
 (4) At equilibrium, $\Delta G = 0$
- 23.** For the reaction

$$\text{F}_2 + 2\text{HCl} \longrightarrow 2\text{HF} + \text{Cl}_2 ;$$

$$\Delta H^0 = -84.4 \text{ K Cal}$$
 If $\Delta H_f^0 (\text{HF}) = -64.2 \text{ K Cal mol}^{-1}$ then $\Delta H_f^0 (\text{HCl})$ is—
 (1) $44.0 \text{ K Cal mol}^{-1}$ (2) $-44.0 \text{ K Cal mol}^{-1}$
 (3) $22 \text{ K Cal mol}^{-1}$ (4) $-22 \text{ K Cal mol}^{-1}$
- 24.** For the reaction $\text{Na}(\text{s}) \longrightarrow \text{Na}(\text{g})$, the heat of reaction is called as
 (1) Heat of vaporisation
 (2) Heat of atomisation
 (3) Heat of sublimation
 (4) Both (2) and (3)
- 25.** Red phosphorus reacts with liquid bromine in exothermic reaction as follows —

$$2\text{P}(\text{s}) + 3\text{Br}_2(\ell) \longrightarrow 2\text{PBr}_3(\text{g})$$

$$\Delta_r H^0 = -243 \text{ KJ}$$
 Enthalpy change when 2.63 g of phosphorus reacts with an excess of bromine is—
 (1) -20.6 KJ (2) -10.3 KJ
 (3) -5.3 KJ (4) -25.6 KJ